

A COUNTEREXAMPLE TO SAMART'S HALF-SHIFT FORMULA FOR n_4

ABSTRACT. Proposition 3.1(iii) of Samart's paper asserts an Eisenstein–Kronecker formula for $n_4(s_4(\tau))$ when $\tau = \frac{1}{2} + iy$ and $y > \frac{1}{2}$. At $\tau_0 = \frac{1}{2} + \frac{\sqrt{6}}{4}i$ we prove, by interval arithmetic, that

$$n_4(s_4(\tau_0)) \geq \frac{151680651}{67108864} > 2.26, \quad \mathcal{E}_4(\tau_0) < 2.22516,$$

where $\mathcal{E}_4$ is the lattice sum in the proposition. Thus the half-shift clause of Proposition 3.1(iii) is false. Since $\text{Im } \tau_0 = \sqrt{6}/4 < 1/\sqrt{2}$, it is that clause alone that is contradicted; the half-shift range was introduced in the 2016 form of the proposition. The proof uses an exact two-dimensional Jensen reduction and a finite interval certificate.

1. THE COUNTEREXAMPLE

For a Laurent polynomial P in r variables, let

$$m(P) = \int_{[0,1]^r} \log \left| P(e^{2\pi i t_1}, \dots, e^{2\pi i t_r}) \right| dt_1 \cdots dt_r$$

and define

$$n_4(s) = 4m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz).$$

The value is independent of the fourth root, by a unit-modulus change of one torus variable.

Write

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n \geq 1} (1 - e^{2\pi i n \tau}),$$

and set

$$A(\tau) = \frac{\eta(\tau)\eta(4\tau)^2}{\eta(2\tau)^3},$$

$$c(\tau) = \left(\frac{\eta(2\tau)}{\eta(\tau)} \right)^6 \left(16A(\tau)^4 + A(\tau)^{-4} \right), \quad s_4(\tau) = c(\tau)^4.$$

This is Samart's parameter $s_4(q)$ with $q = e^{2\pi i \tau}$. For $z \neq 0$, put

$$F(z) = \frac{4(\text{Re } z)^2}{|z|^6} - \frac{1}{|z|^4}, \quad T_d(\tau) = \sum'_{(m,n) \in \mathbb{Z}^2} F(dm\tau + n),$$

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where the prime omits $(0, 0)$, and define

$$\mathcal{E}_4(\tau) = \frac{10 \operatorname{Im} \tau}{\pi^3} (-T_1(\tau) + 4T_2(\tau)).$$

The sums converge absolutely. Samart's Proposition 3.1(iii) states that

$$(1) \quad n_4(s_4(\tau)) = \mathcal{E}_4(\tau)$$

when either $\tau = iy$ with $y \geq 1/\sqrt{2}$, or $\tau = \frac{1}{2} + iy$ with $y > \frac{1}{2}$ [3, Proposition 3.1(iii)].

Theorem 1. *Let*

$$\tau_0 = \frac{2 + \sqrt{-6}}{4} = \frac{1}{2} + \frac{\sqrt{6}}{4}i.$$

Then

$$n_4(s_4(\tau_0)) \geq \frac{151680651}{67108864} > 2.26, \quad \mathcal{E}_4(\tau_0) < 2.22516.$$

Thus (1) fails at a point in the half-shift range of Proposition 3.1(iii).

Although τ_0 also appears in the conjectural Table 6 of [2], the theorem concerns only Proposition 3.1(iii). The half-shift clause is not idle in [3], however: the proof of [3, Thm. 1.2(iii)] in the case $s < 0$ applies it at $(\tau - 1)/2\tau$ with $\tau = iy$, a point whose real part is $\frac{1}{2}$ and whose imaginary part is $1/2y > \frac{1}{2}$, and τ_0 is of exactly this form, with $y = 2/\sqrt{6}$. We do not pursue the consequences for that theorem, or for Table 6, here.

Relation to earlier work. That (1) fails below $\operatorname{Im} \tau = 1/\sqrt{2}$ is not new. Zheng [4, §5] shows that the lattice sum leaves the geometric sheet there and states that below that threshold “the identity fails everywhere”, on numerical evidence along the lines $\operatorname{Re} \tau = 0$ and $\operatorname{Re} \tau = 1/4$, and also reports high-precision numerical evidence that the Table 6 identity $n_4(81) = 40L'(g_7, 0)$ fails. Since $\operatorname{Im} \tau_0 = \sqrt{6}/4 < 1/\sqrt{2}$, that blanket statement already covers τ_0 , so what is new here is not the phenomenon but the target and the mode of proof. Zheng quotes the identity only in its earlier form [1, Prop. 2.1(iii)], for $\operatorname{Im} \tau \geq 1/\sqrt{2}$, and records that below that threshold the identity “is not asserted”. The point of the present note is that it *is* asserted: [3, Prop. 3.1(iii)], which [3] introduces as a “modified version” of [1, Prop. 2.1] and justifies by the remark that the admitted values of τ “lie in a corresponding fundamental domain for $\Gamma_0(N)^*$ ”, covers $\tau = \frac{1}{2} + iy$ for every $y > \frac{1}{2}$, and hence the whole strip $\frac{1}{2} < y < \frac{1}{\sqrt{2}}$. Theorem 1 is thus a proof, rather than numerical evidence, that the formula fails at a point of the range the proposition does assert; membership of such a fundamental domain is not by itself sufficient for it.

2. JENSEN REDUCTION

Lemma 2. *For $c \in \mathbb{C}$, set*

$$a_c(\theta, \phi) = e^{i(\theta+\phi)/2} (e^{i\theta} + e^{i\phi} + c), \quad J(a) = \operatorname{arcosh} \left(\frac{|a-2| + |a+2|}{4} \right),$$

where arcosh is the nonnegative real inverse of $\cosh$. Then

$$(2) \quad n_4(c^4) = \frac{4}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} J(a_c(\theta, \phi)) d\theta d\phi.$$

Proof. The argument of arcosh is at least 1 by the triangle inequality, and $J(-a) = J(a)$. Divide $x^4 + y^4 + z^4 + 1 + cxyz$ by xyz and make the torus substitution

$$u = \frac{x^3}{yz}, \quad v = \frac{y^3}{xz}, \quad w = \frac{z^3}{xy}.$$

Its exponent matrix has determinant 16, so the map is a finite torus covering and preserves normalized Haar measure. Since $uvw = xyz$, the transformed Laurent polynomial is

$$u + v + w + (uvw)^{-1} + c.$$

For fixed $u = e^{i\theta}$ and $v = e^{i\phi}$, Jensen's formula in w reduces the integral to the roots of

$$w^2 + (u + v + c)w + (uv)^{-1}.$$

After a unit-modulus rescaling this is $z^2 + a_c(\theta, \phi)z + 1$; the choice of square root changes a_c to $-a_c$ and hence does not change J .

Let ρ be a root with $|\rho| \geq 1$. Jensen's contribution is $t = \log |\rho|$. Since $-a_c = \rho + \rho^{-1}$,

$$|a_c - 2| + |a_c + 2| = \frac{|\rho + 1|^2 + |\rho - 1|^2}{|\rho|} = 4 \cosh t.$$

Thus the contribution equals $J(a_c)$. The logarithm of a nonzero Laurent polynomial is integrable on the torus, so Fubini's theorem and Jensen's formula apply, proving (2). $\square$

3. VALIDATED BOUNDS

Each inequality below is checked by a finite interval computation from the constants printed here.

The eta parameter. Let $q = e^{2\pi i\tau}$, $r = |q|$, and $R_N = \prod_{n>N} (1 - q^n)$. Then

$$|\log R_N| \leq \sum_{n>N} \frac{r^n}{1 - r^n} \leq \frac{r^{N+1}}{(1 - r)(1 - r^{N+1})},$$

and hence

$$(3) \quad |R_N - 1| \leq \exp \left(\frac{r^{N+1}}{(1 - r)(1 - r^{N+1})} \right) - 1.$$

Using $N = 30$ in the eta products at $\tau_0, 2\tau_0, 4\tau_0$, with the defining factor $e^{\pi i\tau/12}$ evaluated directly from τ , gives

$$(4) \quad 0.88 < \operatorname{Re} c(\tau_0) < 0.89, \quad -0.89 < \operatorname{Im} c(\tau_0) < -0.88.$$

Here $\operatorname{Re} c(\tau_0) = 0.8850840762662\dots$ and $\operatorname{Im} c(\tau_0) = -0.8850840762662\dots$

The lattice sum. The partial-fraction identity

$$\sum_{n \in \mathbb{Z}} (z + n)^{-3} = \frac{\pi^3 \cos(\pi z)}{\sin^3(\pi z)}$$

gives, after pairing the nonzero lattice rows,

$$U_j(\tau) = 2\pi^3 \sum_{m \geq 1} \frac{\cos(j\pi m\tau)}{m \sin^3(j\pi m\tau)},$$

$$\mathcal{E}_4(\tau) = \operatorname{Im} \left(2\pi\tau + \frac{10}{\pi^3} (U_1(\tau) - 2U_2(\tau)) \right).$$

Indeed,

$$\operatorname{Im} \left(\frac{(jm\tau + n)^{-3}}{m} \right) = -j \operatorname{Im}(\tau) F(jm\tau + n),$$

and the omitted $m = 0$ row of each T_j is $6\zeta(4) = \pi^4/15$.

Writing $y = \operatorname{Im} \tau_0$ and $q_j = e^{-2j\pi y}$, the tail after M terms satisfies

$$(5) \quad |U_j - U_{j,M}| \leq \frac{8\pi^3(1 + q_j^{M+1})q_j^{M+1}}{(M+1)(1 - q_j)(1 - q_j^{M+1})^3}.$$

Interval evaluation of the first 20 terms, together with (5), yields

$$(6) \quad \mathcal{E}_4(\tau_0) < 2.22516.$$

The enclosure obtained is $\mathcal{E}_4(\tau_0) = 2.2251552127\dots$, the tail (5) at $M = 20$ being below 10^{-34} .

The Mahler measure. Let

$$c_* = \frac{177}{200}(1 - i), \quad \delta = \frac{123}{3125}.$$

From (4),

$$|c(\tau_0) - c_*| < \frac{\sqrt{2}}{200}, \quad |c(\tau_0)| < \frac{63}{50}.$$

Partition $[0, 2\pi]^2$ into 512^2 equal squares with centers

$$\theta_j = \frac{2\pi(j + 1/2)}{512}, \quad 0 \leq j < 512.$$

Since

$$|\partial_\theta a_c|, |\partial_\phi a_c| \leq 2 + \frac{|c|}{2} \leq \frac{263}{100},$$

every value in the (j, k) -cell lies within

$$\frac{263\pi}{25600} + \frac{\sqrt{2}}{200} < \delta$$

of $a_{c_*}(\theta_j, \theta_k)$. The last inequality follows, for example, from $\pi < 355/113$ and $\sqrt{2} < 99/70$.

Directed interval arithmetic gives lower bounds $\ell_{jk}^\pm$ for $|a_{c_*}(\theta_j, \theta_k) \pm 2|$. With $x_+ = \max\{x, 0\}$, define

$$X_{jk} = \max \left\{ 1, \frac{(\ell_{jk}^- - \delta)_+ + (\ell_{jk}^+ - \delta)_+}{4} \right\},$$

and let r_{jk} be the largest nonnegative integer such that

$$\cosh \left(\frac{r_{jk}}{4096} \right) \leq X_{jk}.$$

A direct interval computation over the 512^2 cells gives the exact sum

$$(7) \quad \sum_{j,k=0}^{511} r_{jk} = 606722604.$$

Of the 262144 cells, 31988 have $X_{jk} = 1$ and hence $r_{jk} = 0$; the other 230156 satisfy $8 \leq r_{jk} \leq 5121$, with $r_{0,0} = 4050$, $r_{255,255} = 1803$, $r_{511,511} = 4080$ and $\sum_k r_{0,k} = 1687754$. No $4096 \operatorname{arcosh}(X_{jk})$ with $r_{jk} > 0$ lies within 10^{-6} of an integer, so the floors are insensitive to the binary64 rounding. By Lemma 2 and monotonicity of arcosh ,

$$n_4(s_4(\tau_0)) \geq \frac{4}{512^2 \cdot 4096} \sum_{j,k} r_{jk} = \frac{151680651}{67108864} > 2.26.$$

Together with (6), this proves Theorem 1.

Reproducibility. The recorded run used Python 3.13.5, NumPy 2.3.5, and mpmath 1.3.0. All transcendental quantities were enclosed with the mpmath interval context at 80 decimal digits. The subsequent binary64 stage assumes IEEE 754 round-to-nearest arithmetic and correctly rounded square root; each result is enlarged outward with `numpy.nextafter`. These semantics form the trusted computing base. Once the cell inequalities are checked, (7) gives the Mahler lower bound as an exact rational number.

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